

Yulin Li

The Hong Kong University of Science and Technology (HKUST)
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RESEARCH INTEREST

Robot Motion Planning & Control, Optimization

EDUCATION

The Hong Kong University of Science and Technology

Hong Kong, China

Ph.D. Student in Robotics and Autonomous Systems

Sep. 2021 – Present

Supervised by **Prof. Jun Ma** and **Prof. Michael Yu Wang**

Harvard University

Cambridge, MA, USA

Visiting Scholar in Computational Robotics Lab

Sep. 2024 – Mar. 2025

Supervised by **Prof. Heng Yang**

University of California, San Diego

San Diego, CA, USA

M.Sc. in Mechanical and Aerospace Engineering GPA: 4.0/4.0

Sep. 2019 – Jun. 2021

Major in: Motion Planning and Control for Robotics

Tongji University

Shanghai, China

B.Eng. in Mechatronic Engineering GPA: 4.7/5.0

Sep. 2015 – Jun. 2019

HONORS & AWARDS

Best Poster Award Finalist, 2024 Workshop on Robot Technology for Sustainable Society

Dec. 2024

Robotica Best Paper Award Finalist, ROBIO 2023

Dec. 2023

Shanghai Excellent Graduated Student

Jun. 2019

Shanghai Scholarship

May. 2018

RoboMasters National College Student Robot Competition 1st Prize Top 9%

May. 2018

Shanghai College Student Mechanical Engineering Innovation Competition 1st Prize

Apr. 2017

China Undergraduate Mathematical Contest in Modeling 1st Prize Top 0.82%

Oct. 2017

RESEARCH & INTERNSHIP

HKUST: Robot Motion Planning and Control Lab

- Advised by **Professor Jun Ma** and **Professor Michael Yu Wang**
- Developing geometry-aware safety-critical controller and trajectory optimization algorithms for robot navigation in unknown and complex environments.
- In charge of the research development of the motion planning & control system of a mobile manipulator to achieve safe motion and dexterous manipulation in cluttered indoor environments.

Harvard University: Computational Robotics Lab

- Advised by **Professor Heng Yang**
- Conducting research on contact-implicit motion planning problems, developing high-performance numerical solvers for nonlinear programming with complementarity constraints.

Tencent Holding Ltd: Robotics X Lab

- Advised by **Professor Zhengyou Zhang**
- Build a single leg hopping platform to conduct experiments on new designs of the actuator and leg structure.

Carnegie Mellon University Robotics Institute

- Advised by **Professor Howie Choset**
- Carried out snake robot motion planning and adaptive control.

- [1] **Y. Li**, H. Han, S. Kang, J. Ma, H. Yang, “On the Surprising Robustness of Sequential Convex Optimization for Contact-Implicit Motion Planning,” *Robotics: Science and Systems (RSS) 2025, submitted*.
- [2] **Y. Li**, C. Zheng, K. Chen, Y. Xie, X. Tang, M. Y. Wang, and J. Ma, “Collision-Free Trajectory Optimization in Cluttered Environments with Sums-of-Squares Programming,” *IEEE Robotics and Automation Letters, 2024*.
- [3] **Y. Li**, X. Tang, K. Chen, C. Zheng, H. Liu, and J. Ma, “Geometry-Aware Safety-Critical Local Reactive Controller for Robot Navigation in Unknown and Cluttered Environments,” *IEEE Robotics and Automation Letters, 2024*.
- [4] **Y. Li**, Z. Song, C. Zheng, Z. Bi, K. Chen, M. Y. Wang and J. Ma, “FRTree Planner: Robot Navigation in Cluttered and Unknown Environments with Tree of Free Regions,” *IEEE Robotics and Automation Letters, 2025*.
- [5] C. Zheng, **Y. Li** (co-first), Z. Song, Z. Bi, J. Zhou, B. Zhou, J. Ma, ‘Local Reactive Control for Mobile Manipulators with Whole-Body Safety in Complex Environments,’ *IEEE Robotics and Automation Letters, 2025*.
- [6] K. Chen, H. Liu, **Y. Li**, J. Duan, L. Zhu, and J. Ma, “Robot navigation in unknown and cluttered workspace with dynamical system modulation in starshaped roadmap,” *IEEE International Conference on Robotics and Automation (ICRA), 2025*.
- [7] Y. Wang, **Y. Li**, Z. Peng, H. Ghazzai, and J. Ma, “Chance-Aware Lane Change with High-Level Model Predictive Control Through Curriculum Reinforcement Learning,” *IEEE International Conference on Robotics and Automation (ICRA), 2024*.
- [8] H. Liu, Z. Huang, Z. Zhu, **Y. Li**, S. Shen, and J. Ma, “Improved Consensus ADMM for Cooperative Motion Planning of Large-Scale Connected Autonomous Vehicles with Limited Communications,” *IEEE Transactions on Intelligent Vehicles, 2024*.
- [9] Z. Cheng, **Y. Li** (co-first), K. Chen, J. Duan, J. Ma, and T. H. Lee, “Neural-iLQR: A Learning-Aided Shooting Method for Trajectory Optimization,” *IEEE International Conference on Robotics and Biomimetics (ROBIO), 2023*.
- [10] H. Liu, K. Chen, **Y. Li**, Z. Huang, J. Duan, and J. Ma, “Integrated Decision-Making and Control for Urban Autonomous Driving with Traffic Rules Compliance,” *IEEE International Conference on Robotics and Biomimetics (ROBIO), 2023*.
- [11] K. Chen, J. Cao, **Y. Li**, J. Ma, “G2-SDF: Geometry-Guided Neural Signed Distance Fields for Scalable and Detailed Reconstruction,” *IEEE Robotics and Automation Letters, submitted*
- [12] J. Cao, Q. Zhang, J. Sun, J. Wang, H. Cheng, **Y. Li**, J. Ma, Yecheng Shao, Wen Zhao, G. Han, Y. Guo, R. Xu, “Mamba Policy: Towards Efficient 3D Diffusion Policy with Hybrid Selective State Models,” *International Conference on Intelligent Robots and Systems (IROS), 2025, submitted*
- [13] Z. Bi, K. Chen, C. Zheng, **Y. Li**, H. Li, J. Ma, “Interactive Navigation for Legged Manipulators with Learned Arm-Pushing Controller,” *International Conference on Intelligent Robots and Systems (IROS) 2025, submitted*
- [14] H. Liu, K. Chen, **Y. Li**, Z. Huang, and J. Ma, “UDMC: Unified Decision-Making and Control Framework for Urban Autonomous Driving with Motion Prediction of Traffic Participants,” *IEEE Transactions on Intelligent Transportation Systems, submitted*.

REFERENCES

Prof. Jun Ma

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Prof. Heng Yang

School of Engineering and Applied Sciences
Harvard University
Email: hankyang@seas.harvard.edu

Prof. Michael Yu Wang

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Great Bay University
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Prof. Xindong Tang

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Hong Kong Baptist University
Email: xdtang@hkbu.edu.hk